

Co3-AT2

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A shepherd boy gets bored tending the town's flock. To have some fun he cries out wolf! even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

one night, the shepherd boy sees a real wolf approaching the flock.

The villagers refuse to be fooled again and stay in their houses.

Identify TP, FP, FN, TN.

Act \ pred		
	Y	N
Y	TP	wolf came FN
N	NO wolf FP	TN

Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of ~~cat~~ 30 cat images. It also predicts absence of a cat in 50.

Total = 100

Actual cat images = 30

A \ P	P	
	Y	N
P'	25	20
	5	50
	25	70

$$\text{precision} = \frac{25}{30} \times 100$$

$$= 83.3$$

$$\text{recall} = \frac{25}{25+20} = \frac{25}{45} \times 100 = 55.5$$

$$F = \frac{2 \times 0.83 \times 0.55}{0.83 + 0.55} = \frac{0.913}{1.38} \times 100$$

$$= 66.159.$$

An email service provider wants to classify emails as spam or not spam using features.

③ contains "offer" (yes/no)

contains "free" (yes/no)

Email length (short/long)

sample data set

offer free length class

yes yes short spam

yes no long not spam

no yes short spam

no no long not spam

yes yes long spam

no no short not spam.

① calculate prior probabilities. $P(\text{spam})$, $P(\text{not spam})$

$$P(\text{spam}) = \frac{\text{Number of spam Emails}}{\text{Total Emails}}$$

$$= \frac{3}{6} = \frac{1}{2} = 0.5$$

Not spam

$$\frac{3}{6} = \frac{1}{2} = 0.5$$

⑤ conditional probabilities for each ~~feature~~ feature.

offer	free	length	class
yes	yes	short	spam
yes	no	long	Not spam
no	yes	short	spam
no	no	long	Not spam
yes	yes	long	spam
no	no	short	not spam.

offer = $\frac{2}{3} \leftarrow$ spam $\frac{1}{3} \leftarrow$ NO spam.

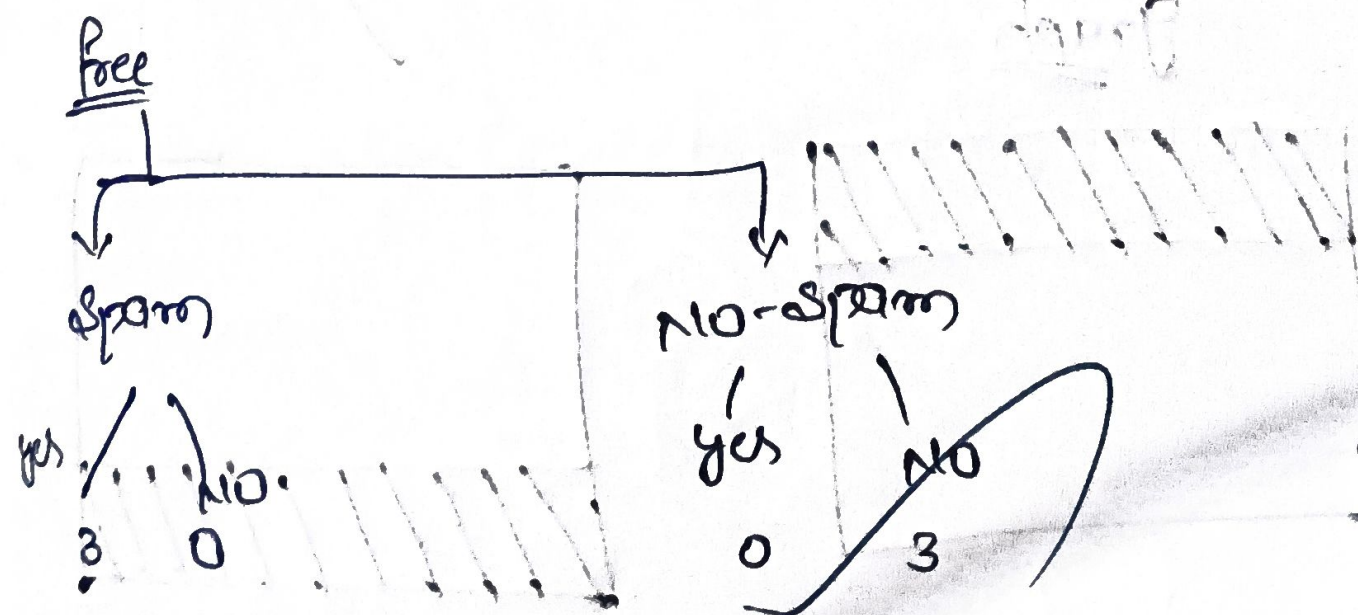
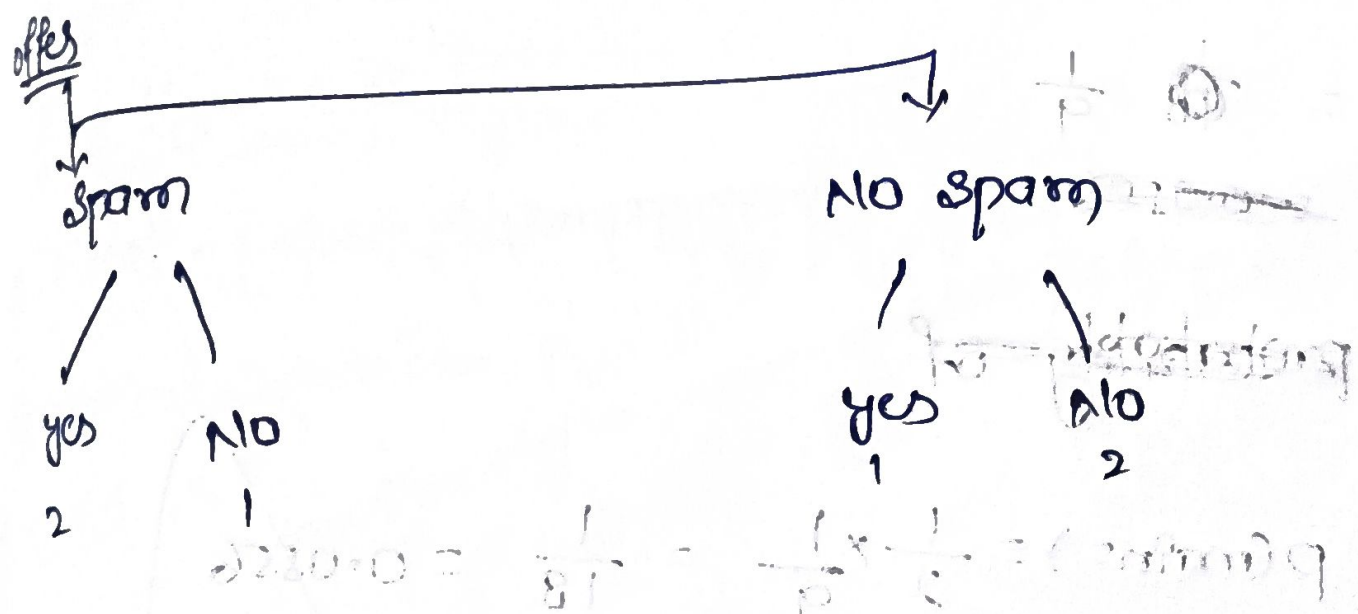
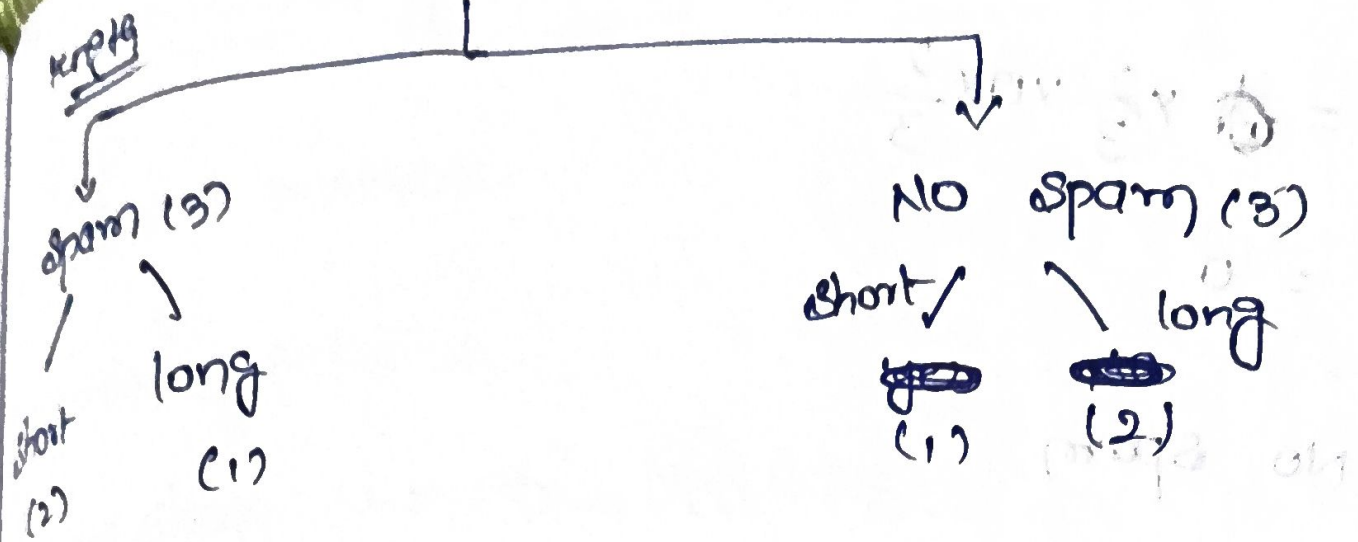
free = $\frac{3}{3} = 1 \leftarrow$ spam $\frac{0}{3} = 0 \leftarrow$ NO spam.

length.

$$p(\text{length} = \frac{\text{short}}{\text{spam}}) = \frac{2}{3}$$

$$p(\text{length} = \frac{\text{long}}{\text{spam}}) = \frac{1}{3}$$

$$\begin{array}{r} 3 \overline{) 1.00} \\ \underline{3} \\ 0 \\ \underline{0} \\ 0 \\ \underline{0} \\ 0 \end{array}$$



② classify the New Email.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

offer = yes, free = NO, length = short

$$= \cancel{1} \times \frac{2}{3} \times 0 \times \frac{2}{3}$$

$$= 0$$

No spam

$$= \cancel{1} \times \frac{1}{3} \times 1 \times \frac{1}{3}$$

$$= \cancel{1} \times \frac{1}{9}$$

~~probability~~

~~probability of~~

$$P(\text{main}) = \frac{1}{2} \times \frac{1}{9} = \frac{1}{18} = 0.0556$$

graph

